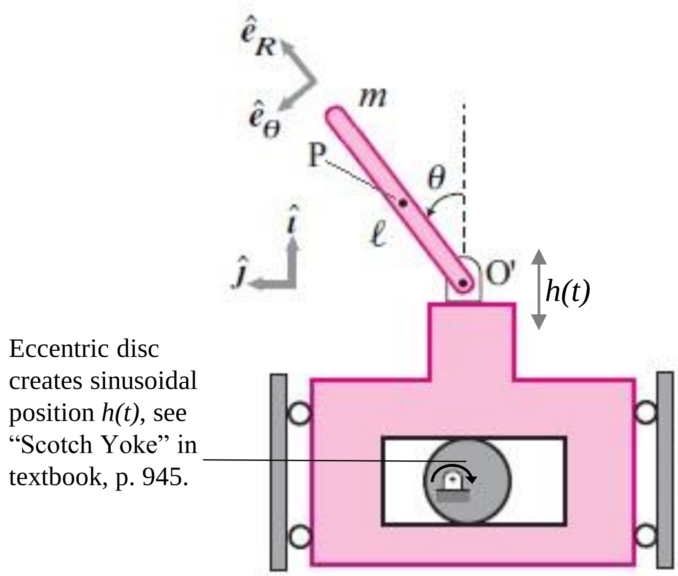
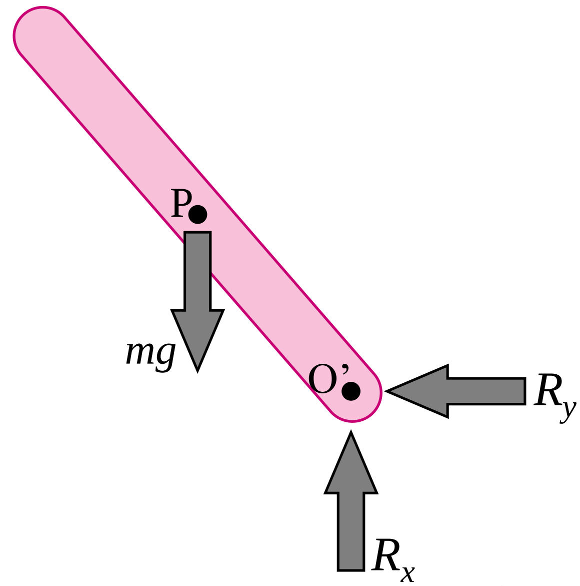
24. Inverted pendulum with shaking base. Some stick of length ℓ is attached to a hinge at its bottom, the position of the hinge *h(t)* varies sinusoidally with time.

<u>(a) Find equations of motion.</u>



System Diagram

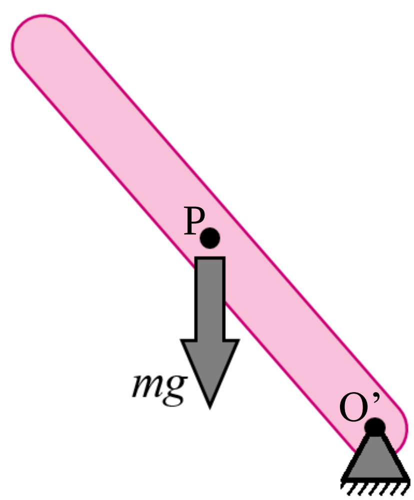


System Free 'Body' Diagram

Method 1: AMB about point O'.

O' moves with pre-specified acceleration, *F* moves at same rate as O', can circumvent constraint forces.

$$\dot{H}_{/O'} = \vec{r}_{P/O'} \times \sum_i \vec{F}_i = \vec{r}_{P/O'} \times \vec{a}_{P/F} + I^P\ddot{\theta}$$



Writing out terms:

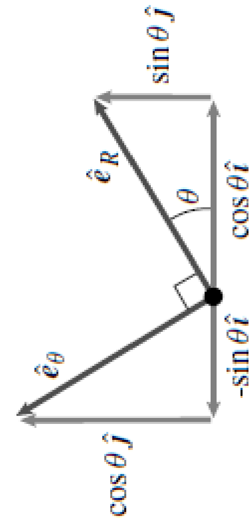
$$\sum_i \vec{F}_i = -mg\hat{\imath}$$

$$\vec{r}_{P/O'} = \frac{\ell}{2}(cos\theta\hat{\imath} + sin\theta\hat{\jmath}) = \frac{\ell}{2}\hat{e}_r$$

Continued on next page.

$\vec{a}_{P/F} = \vec{a}_{P/O'} + \vec{a}_{O'}$

$\ddot{\vec{r}}_{P/O'} = \vec{a}_{P/O'} = -\frac{\ell}{2}\dot{\theta}^2\hat{e}_r + \frac{\ell}{2}\ddot{\theta}\hat{e}_\theta$



Taken from textbook p. 670. Rotated to fit problem.

Given $\vec{a}_{O'} = a(t)$:

$\vec{a}_{P/F} = -\frac{\ell}{2}\dot{\theta}^2\hat{e}_r + \frac{\ell}{2}\ddot{\theta}\hat{e}_\theta + a(t)\hat{\imath}$

Right hand rule entails that:
$\hat{e}_r \times \hat{e}_r = 0,\ \hat{e}_r \times \hat{e}_\theta = \hat{k},\quad \hat{e}_r \times \hat{\imath} = -sin\theta\hat{k},\quad \hat{e}_r \times \hat{\jmath} = cos\theta\hat{k}$
(Check to make sure!)

$$mg\frac{\ell}{2}sin\theta\hat{k} = m\frac{\ell}{2}\left(\frac{\ell}{2}\ddot{\theta} - sin\theta a(t)\right)\hat{k} + I^P\ddot{\theta}\hat{k}$$

For h(t) $= dsin(\omega_o t),\ a(t) = -\mathrm{d}\omega_o^2 sin\omega_o,\ I^P = \frac{m\ell^2}{12}$:

$$gsin\theta = \frac{\ell}{2}\ddot{\theta} + sin\theta\mathrm{d}\omega_o^2 sin(\omega_o t) + \frac{\ell}{6}\ddot{\theta}$$

$$\boxed{\frac{2\ell}{3}\ddot{\theta} = sin\theta(\mathrm{g} - \mathrm{d}\omega_o^2 sin\omega_o)}$$

Method 2: LMB and AMB of whole system.

Obtain accelerations of x and y components of system:

$$x = \frac{\ell}{2}\cos\theta + h(t) \qquad y = \frac{\ell}{2}\sin\theta$$

$$\dot{x} = -\frac{\ell}{2}\dot{\theta}\sin\theta + v(t) \qquad \dot{y} = \frac{\ell}{2}\dot{\theta}\cos\theta$$

$$\ddot{x} = \frac{\ell}{2}\left(-\dot{\theta}^2\cos\theta - \ddot{\theta}\sin\theta\right) + a(t) \qquad \ddot{y} = \frac{\ell}{2}\left(-\dot{\theta}^2 sin\theta + \ddot{\theta}cos\theta\right)$$

NOTE: x and y coordinates are measured from a “non-moving” point O to point P, thus includes OO’ + O’P (see system diagram).

Perform Linear Momentum Balance on body:

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} \Rightarrow \ddot{\vec{r}} = \ddot{x}\hat{\imath} + \ddot{y}\hat{\jmath}$$

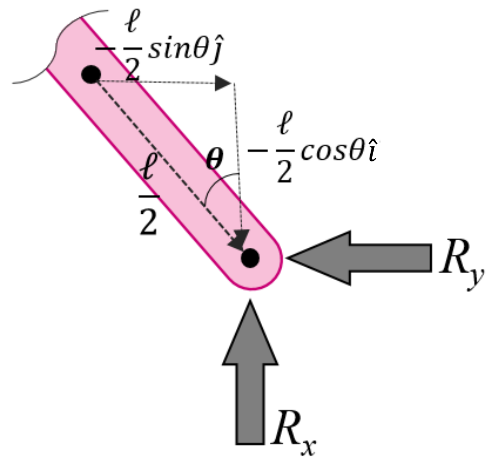
$$\sum_{all}\vec{F} = m\ddot{\vec{r}} = (R_x - mg)\hat{\imath} + R_y\hat{\jmath}$$

Use constraint force-angular relationships and AMB to eliminate unknowns:

Find R_x and R_y in from LMB:

$$R_x = m\left[\frac{\ell}{2}\left(-\dot{\theta}^2\cos\theta - \ddot{\theta}\sin\theta\right) + a(t)\right] + mg \qquad R_y = m\frac{\ell}{2}\left(-\dot{\theta}^2 sin\theta + \ddot{\theta}cos\theta\right)$$

Then, perform AMB about the center of mass:



$$\sum_{all}\dot{\vec{H}} = \sum_i \vec{r}_i' \times \vec{F}_i = \vec{r}_i' \times \left(R_x\hat{\imath} + R_y\hat{\jmath}\right)$$

$$I^P\ddot{\theta}\hat{k} = -\frac{\ell}{2}cos\theta R_y(\hat{\imath}\times\hat{\jmath}) - \frac{\ell}{2}sin\theta R_x(\hat{\jmath}\times\hat{\imath})$$

Algebra and trig identities; $\dot{\theta}^2$ drops out, $\ddot{\theta}$ terms combine.

$$\frac{m\ell^2}{12}\ddot{\theta} = -m\frac{\ell^2}{4}\ddot{\theta} + m\frac{\ell}{2}sin\theta(a(t)+g)$$

$$\frac{2\ell}{3}\ddot{\theta} = sin\theta(a(t)+g)$$

$$\boxed{\frac{2\ell}{3}\ddot{\theta} = sin\theta(\mathrm{g} - \mathrm{d}\omega_o^2\sin(\omega_o t))}$$

NOTE: Method 2 takes longer than Method 1 but prepares us for DAE better than Method 1.

Forming DAE:

Given x, y, R_x, R_y, θ, need five equations to solve linear system. LMB, AMB give three equations, 2 equations needed to constrain bar to pin.

$$\text{LMB, AMB}\begin{cases} R_x - mg = m\ddot{x} \\ R_y = m\ddot{y} \\ -\frac{\ell}{2}cos\theta R_y + \frac{\ell}{2}sin\theta R_x = \frac{m\ell^2}{12}\ddot{\theta} \end{cases}$$

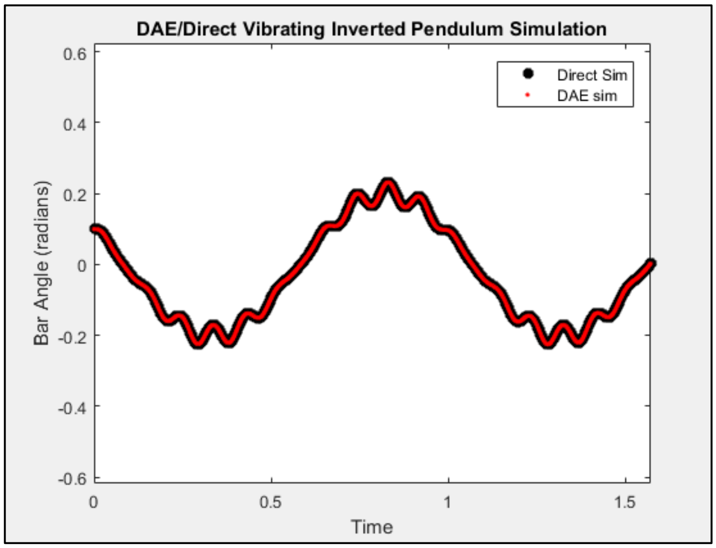
$$\text{Constraint Eqn's}\begin{cases} \ddot{x} = \frac{\ell}{2}\left(-\dot{\theta}^2\cos\theta - \ddot{\theta}\sin\theta\right) - \mathrm{d}\omega_o^2 sin(\omega_o t) \\ \ddot{y} = \frac{\ell}{2}\left(-\dot{\theta}^2 sin\theta + \ddot{\theta}cos\theta\right) \end{cases}$$

Matrix will be shown in part (c).

(b) Show stability of pendulum from simulating equations.

Code for parameters and direct simulation

```
%Ronald Heisser
clear all

s.g = 9.8;          %
s.l = 1;            %Define parameters
s.d = .1;           %
s.m = 1;            %
s.omega = 70;       %

initial = [.1 0];%initial conditions

opts = odeset('reltol',1e-9,'abstol',1e-9);          %
[t, data1] = ode45(@rhs, [0 5],initial,opts,s);      %ODE code
[t2, data2] = ode45(@rhsdae, [0 5],initial,opts,s);%

plot(t,data1(:,1),'k*',t2,data2(:,1),'r.','LineWidth',3)  %Plotting
legend('Direct Sim','DAE sim')                             %
title('DAE/Direct Vibrating Inverted Pendulum Simulation')%
xlabel('Time')                                             %
ylabel('Bar Angle (radians)')                              %
axis equal                                                 %

function zdot = rhs(t,z,s)
%This is direct numerical simulation (Method 1)
apos = -s.d*s.omega^2*sin(s.omega*t);

zdot(1) = z(2);
zdot(2) = 3/(2*s.l)*sin(z(1))*(s.g+apos);
%y(1) is theta, y(2) is theta dot
zdot = zdot';

end
```



Solutions check out. Plot includes DAE Simulation. See last page for part (c) and its related code.

(c) Do it again for DAE.

$$\begin{bmatrix} 1 & 0 & 0 & -m & 0 \\ 0 & 1 & 0 & 0 & -m \\ -\ell/2\, sin\theta & \ell/2\, cos\theta & m\ell^2/12 & 0 & 0 \\ 0 & 0 & \ell/2\, sin\theta & 1 & 0 \\ 0 & 0 & \ell/2\, cos\theta & 0 & 1 \end{bmatrix} \begin{bmatrix} R_x \\ R_y \\ \ddot{\theta} \\ \ddot{x} \\ \ddot{y} \end{bmatrix} = \begin{bmatrix} mg \\ 0 \\ 0 \\ -\ell/2\, \dot{\theta}^2 cos\theta + a(t) \\ -\ell/2\, \dot{\theta}^2 sin\theta \end{bmatrix}$$

Where *a(t)* is $-\mathrm{d}\omega_o^2 \sin(\omega_o t)$.

```
function ydot = rhsdae(t,y,s)
%This is DAE numerical simulation (Method 2)
theta = y(1);                             %
thdot = y(2);                             %
c1 = s.l/2;                               %Define Parameters
c2 = s.m*s.l^2/12;%matrix shortcuts       %
accel = s.d*s.omega^2*sin(s.omega*t); %

%DAE Matrix, done a row at a time
E = zeros(5,5);
E(1,:) = [  1                   0             0             -s.m     0];
E(2,:) = [  0                   1             0              0    -s.m];
E(3,:) = [  -c1*sin(theta)  c1*cos(theta)     c2             0       0];
E(4,:) = [  0                   0         c1*sin(theta)      1       0];
E(5,:) = [  0                   0        -c1*cos(theta)      0       1];

%RHS vector
fterms = [s.m*s.g...%R_x row
          0 ...      %R_y row
          0 ...       %theta ddot row
          -c1*thdot^2*cos(theta)-accel...%xddot row
          -c1*thdot^2*sin(theta)]';    %yddot row

X = E\fterms;

ydot(1) = y(2); %move thetadot to theta
ydot(2) = X(3); %move thetaddot to thetadot

ydot = ydot';

end
```